

Experiment # 04

Building Simple Network Topologies

Objective:

In this lab we will learn about network topologies.

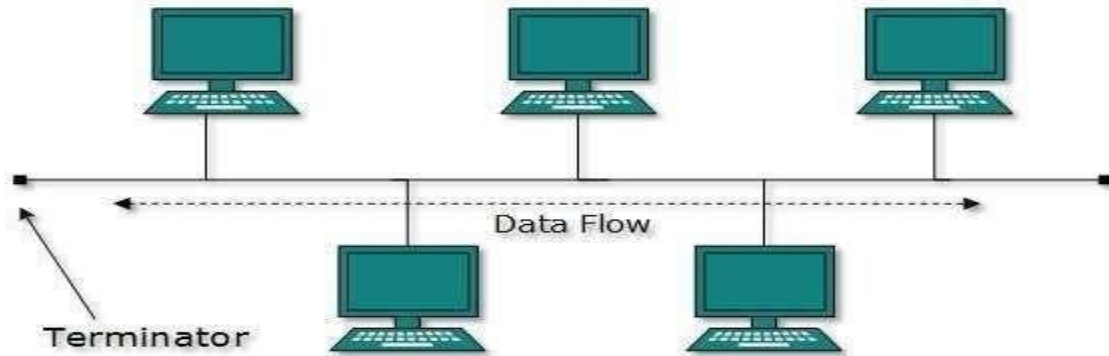
Software Tool:

- Cisco Packet Tracer

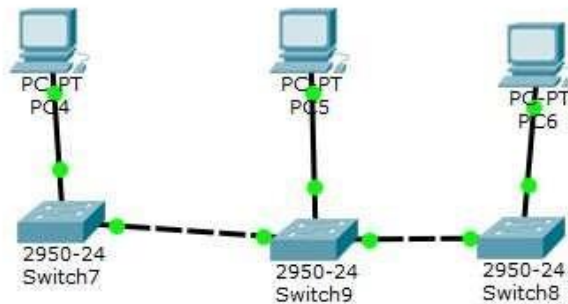
Theory:

Network topology is the arrangement of the various elements of a computer network.

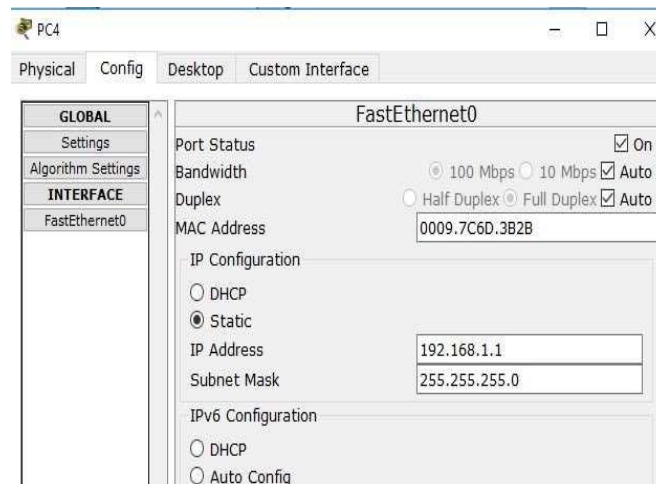
Bus Topology: All devices are connected to a central cable, called the bus or backbone. Bus networks are relatively inexpensive and easy to install for small networks.



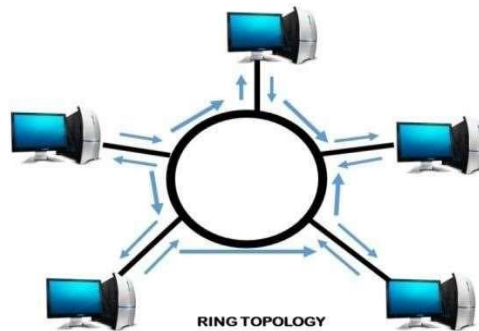
Connect three PC's with switches and join them using switches.



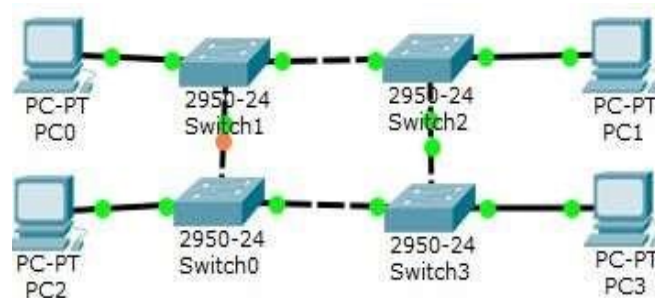
Configure IP's from 198.168.1.1 to 198.168.1.3.



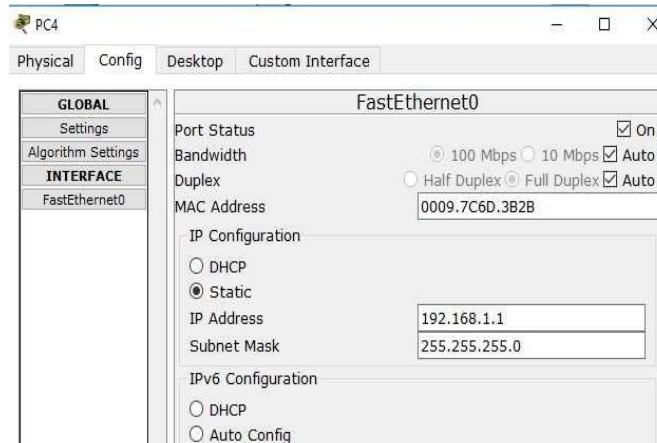
Ring Topology: All devices are connected to one another in the shape of a closed loop, so that each device is connected directly to two other devices, one on either side of it. Ring topologies are relatively expensive and difficult to install, but they offer high bandwidth and can span large distances.



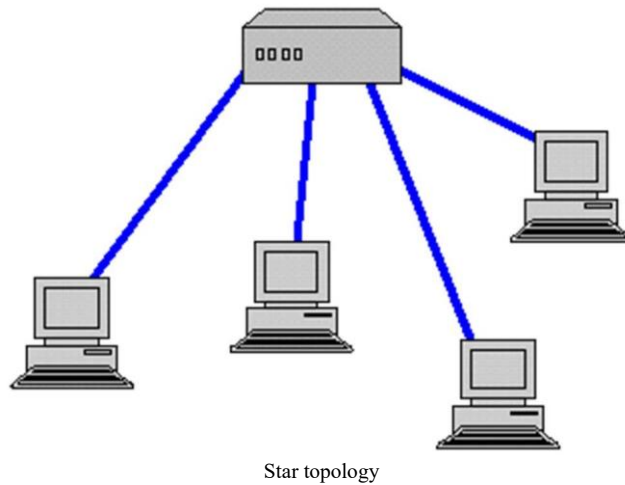
Connect 4 PC's to switches connect every PC's to a switch and then connect last switch to first switch as shown in the figure.



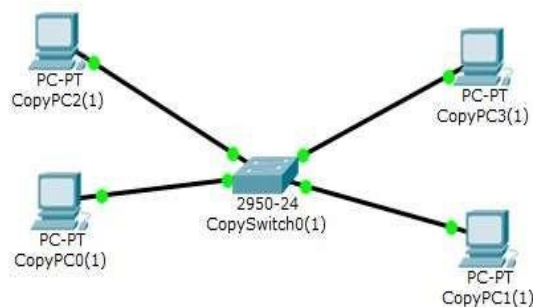
Configure IP's from 192.168.1.1 to 198.161.1.4.



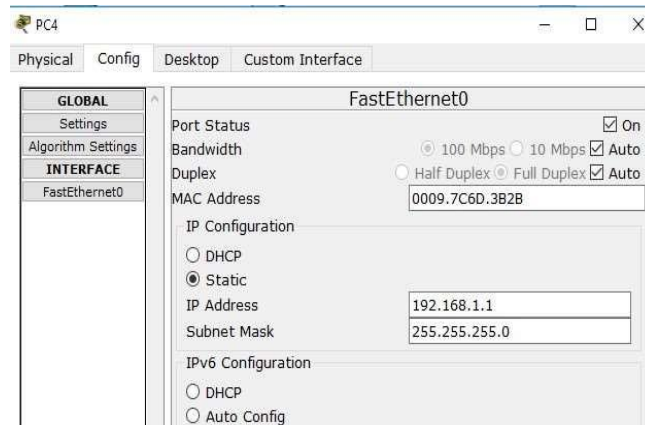
Star Topology: A star topology is a topology for a Local Area Network (LAN) in which all nodes are individually connected to a central connection point, like a hub or a switch. A star takes more cable than e.g. a bus, but the benefit is that if a cable fails, only one node will be brought down.



Connect every PC's with switch

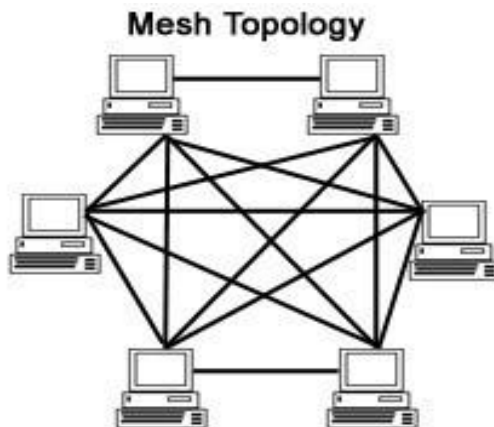


Configure IP's from 198.161.1.1 to 198.161.1.4

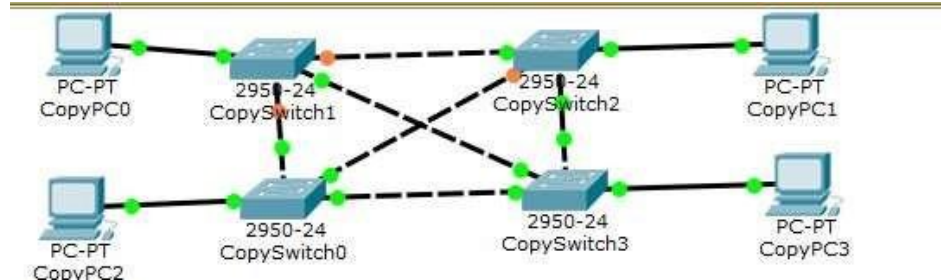


Mesh Topology: A network setup where each computer and network device is interconnected with one another, allowing for most transmissions to be distributed, even if one of the connections go down. This topology is not commonly used for most computer networks as it is difficult and expensive to have redundant connection to every computer. However, this topology is commonly used for wireless networks.

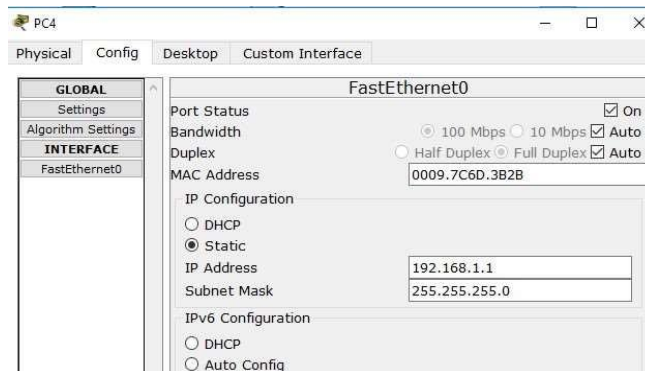
Below is a visual example of a simple computer setup on a network using a mesh topology.



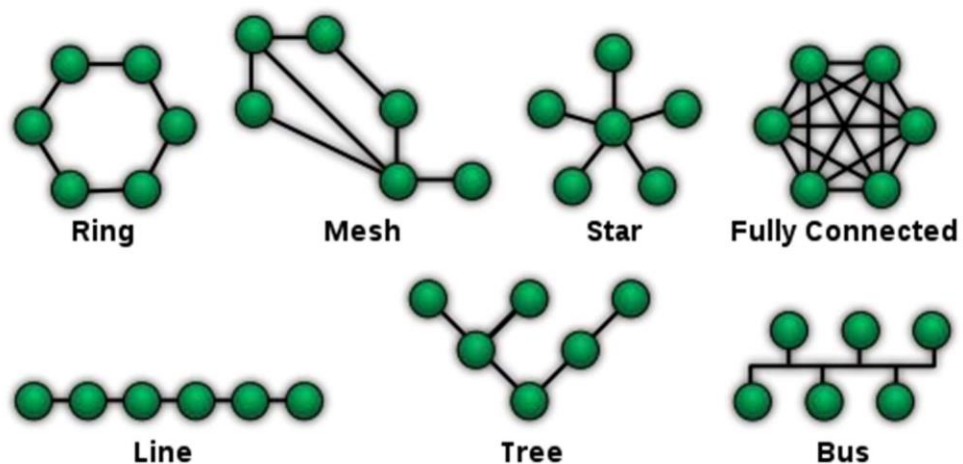
Connect 4 PC's with switch and connect every all switches with each other.



Configure IP's from 198.161.1.1 to 198.161.1.4

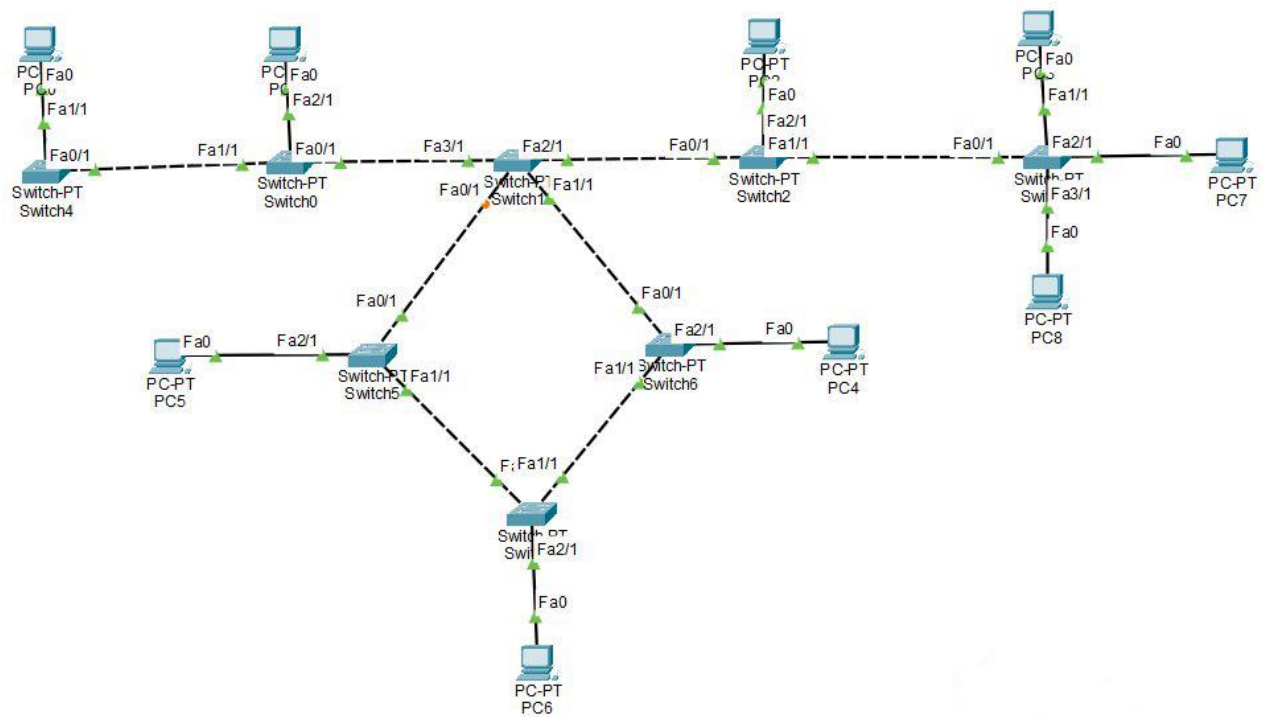


Hybrid Topology: A topology which contains more than one of the upper topologies or it is the mixture of the above topologies.



LAB TASK

1. Practice all the Topologies of Packet Tracer.
2. Simulate a network which contains 9 PCs connected using Hybrid topology. Assign IP addresses to every PC from 192.168.1.1 to 192.168.1.9.



Conclusion
